

Goal

Create a Windows PC application that monitors an AV device. The application should be capable of the following.

Requirements

1. Establish Ethernet Connectivity

Connect to a standard AV device* such as a DSP, display, control processor, or AV transmitter/receiver. Supported connection protocols include TCP, UDP, WebSocket, RESTful API, SSH, or Telnet.

* In the case where you do not have access to an AV device, you may instead use a public API to mimic this functionality with an their endpoints and services. You'd also then be responsible for translating the requirements into similar functionality based on that API's capabilities and responses.

2. Device Heartbeat Monitoring

Implement a heartbeat mechanism to check the device's status every 5 minutes while the application is running. Display the device's status as "Online" if a response is received, and "Offline" otherwise. The status can be shown via a user interface (UI) or console.

3. Retrieve Device Configuration on Startup

At application startup, retrieve and display key device configuration details, including IP address, MAC address, and serial number, in the UI or console.

4. Real-Time Status Monitoring

Continuously monitor the device's real-time status. For example, track the power state of a display or the mute block status of a DSP.

Suggestions

If you're not going to use AV hardware and instead going to use an API, here are some suggestions

- List of public APIs: <https://www.postman.com/cs-demo/public-rest-apis/collection/tfzpqfc/public-rest-apis>
- Public API I'd suggest: <https://haveibeenpwned.com/API/v2>

Since this will be a Windows PC application, I'd recommend using Visual Studio Community edition. The type of application is up to you; web app, console app, console app – your call.

C# is the preferred language for this exercise, and we will want to review the code after the exercise.